# EXOPLANET HUNTERS(stac wars)

STAC Wars, Astronomy Task (PS3: Exoplanet Hunters)

**Abstract:** Relative-brightness time series of three stars (STAC-1, STAC-2, STAC-3) were analysed for transit signals. Each star shows repeating, equally depthed dips, indicating one transiting planet per star. The estimated orbital periods are approximately 4, 80 and 40 days, with transit depths of 0.0030, 0.0010 and 0.00088. The derived planet radii are 5.98, 2.76 and 1.62 Earth radii, and the orbital distances are 0.049, 0.337 and 0.182 AU. STAC-3 hosts the most promising candidate: it is the smallest planet (a Super-Earth) and lies closest to its star's habitable-zone distance (64%).

Index Terms: exoplanets, transit photometry, Kepler's third law, habitable zone

## *INTRODUCTION* ( )

When a planet passes in front of its host star, it blocks a small fraction of the light and the star appears slightly dimmer. A single dip could be glitch, starspot , but dips that repeat at equal intervals imply an object on a repeating orbit, which is the significance of a planet . The depth of each dip gives the planet's size relative to the star, and the time between dips gives the orbital period. With the star's mass, Kepler's third law then gives the orbital distance. This report applies these ideas to three stars and ranks the detected planets by their potential for further habitability studies .

## I. DATA ASSUMPTIONS AND METHODS

### *A. Data.*

Stellar properties are given in Table 1.Brightness was recorded every 1 d for STAC-1(days 0-20), every 20 d for STAC-2 (days 0-200) and every 10 d for STAC-3 (days 0-200)

TABLE 1. STELLAR PROPERTIES

| | STAC-1 | STAC-2 | STAC-3 |
|---|---|---|---|
| Mass($M_\odot$) | 1.00 | 0.80 | 0.50 |
| Radius($R_\odot$) | 1.00 | 0.80 | 0.50 |
| Luminosity ($L_\odot$) | 1.00 | 0.50 | 0.08 |
| Temperature( $T_\odot$) | 5800 | 5000 | 3800 |

### *B. Assumptions*

(1) Each star has one transiting planet on a circular orbit.

(2) Brightness dips are caused only by the planet.

(3) The orbital period is estimated from the spacing of the dips; where spacings vary, the most common spacing is adopted and the effect is tested in Section IV.

(4) Where several dips differ slightly in depth, their mean is used.

(5) Size classes: Earth-sized 0.8–1.25 $R_\oplus$, Super-Earth 1.25–2 $R_\oplus$, Neptune-sized 3–6 $R_\oplus$, Jupiter-sized 9 $R_\oplus$ and above; a planet between classes is assigned to the nearest one.

(6) $R_\odot$ = 696,000km, $R_\oplus$ = 6,371 km, 1 AU = 149.6 million km, 1 yr = 365.25 d

### *C. Equations. The transit depth is*

$$\delta = (F_{normal} - F_{transit}) / F_{normal}. \quad (1)$$

The planet's disc blocks a fraction of the star's disc equal to the ratio of their areas, so

$$\delta \approx (R_p / R_s)^2, \text{ so } R_p = R\sqrt{\delta} \quad (2)$$

Kepler's third law in solar units, with P in years, a in AU and M in solar masses, is

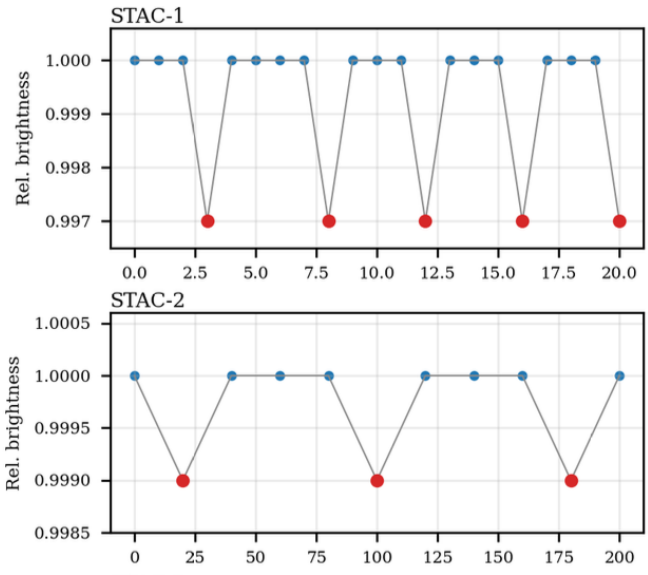
$$P^2 = a^3 / M, \text{ so } a = (M P^2)1/3 \quad (3)$$

The habitable-zone distance and the comparison ratio are

$$dHZ \approx \sqrt{(L / L_\odot)} \text{ AU, ratio} = a / dHZ \quad (4)$$

## *II.* RESULT

A. Detection (Q1–Q3). All three stars show a periodic decrease in brightness (Fig. 1), so each indicates a transiting planet..



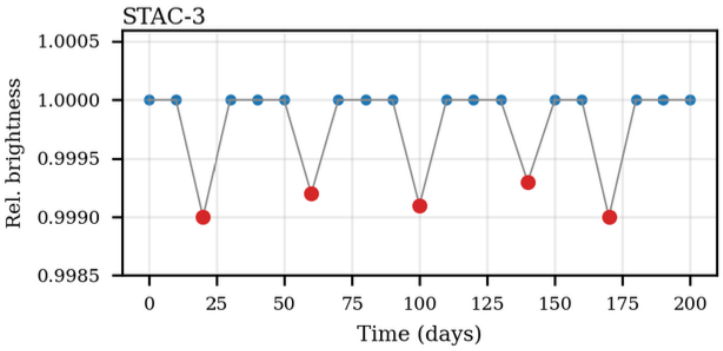
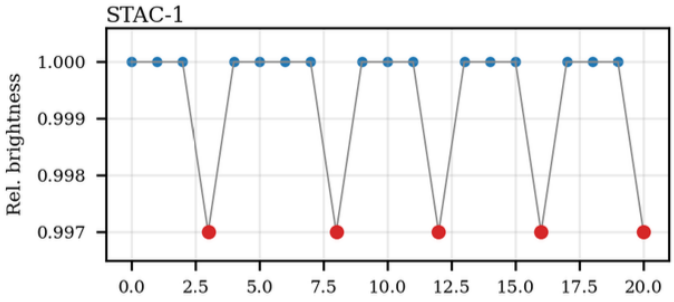



*Fig. 1. Light curves of the three stars. Red points mark in-transit readings.*

# STAC-1

Useful Constants
The following astrophysical constants and conversions are utilized for this analysis:
- Solar radius, $R_\odot$ = 696, 000 km
- Earth radius, $R_\oplus$ = 6371 km
- 1 AU = 149.6 × 106 km
- 1 year = 365.25 days
- $1M_\odot$ = mass of the sun
- $1L_\odot$ = luminosity of the sun
- Mass: $1.00M_\odot$
- Radius: $1.00R_\odot$
- Luminosity: $1.00L_\odot$
- Surface temperature: 5800 K




## Observations and Planetary Radius

Brightness observations indicate transit dips at days 3, 8, 12, 16, and 20. The orbital period (P ) can be approximated as 4 days:

$$P=4/365.25 \text{ yr} \quad (1)$$

The transit depth (δ) is calculated as follows:

$$\delta=1.0000- 0.9970/1.000 = 0.003 \quad (2)$$

The ratio of the planetary radius ($R_p$) to the stellar radius ($R_s$) is derived directly from the transit depth:

$$(R_p/ R_s)^2 = \delta \quad (3)$$

$$R_p/R_s = \sqrt{\delta} = \sqrt{0.003} = 0.055 \quad (4)$$

Given $R_s = 1.00R_\odot$ = 696, 000 km, the planetary radius is:

$$R_p = 0.055 \times 696, 000 = 38, 280 \text{ km} \quad (5)$$

Converting this measurement to Earth radii ($R_\oplus$):

$$38280/6371 = 6.008R_\oplus \quad (6)$$

Based on standard exoplanet classifications, this radius corresponds to a Neptune-sized planet:
- Earth size:~ $0.8R_\oplus$ to $1.25R_\oplus$
- Super-Earth:~ $1.25R_\oplus$ to $2R_\oplus$
- Neptune-sized:~ 3 to $6R_\oplus$
- Jupiter-sized:~ $9R_\oplus$ and up

## Orbital Distance and Habitability

Using Kepler's Third Law, the semi-major axis (a) is calculated:

$$P^2 = a^3 /M \quad (7)$$

$$a = (P^2 \times M )^{1/3} \quad (8)$$

$$a =((4/365.25)^2\times 1)^{1/3} \approx 0.05 \text{ AU} \quad (9)$$

a=0.05*149.6million km

=7.48million years

Converting to kilometers:

$$a = 0.05 \times 149.6 \times 10^6 = 7.48 \times 10^6 \text{ km} \quad (10)$$

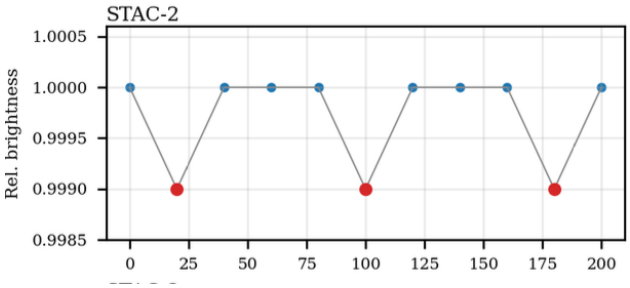
So this planet id about 20 times closer to its star than earth is to the sun .

$$dHZ = \sqrt{1} = 1 \text{ AU} \quad (11)$$

- This planet sits at 0.05AU , only about 5% of that . Far too hot and it is Neptune-sized ,so it is a poor home.

# STAC-2

Second host star properties :
- Mass: 0.80M⊙
- Radius: 0.80R⊙
- Luminosity: 0.40L⊙
- Surface temperature: 5000 K




## Observations and Planetary Radius:

Brightness observations indicate transit dips at days 20, 100, and 180, resulting in a gap of 80 days. The orbital period (P ) is calculated as:
P=80/365.25 yr = 0.22 yr (12)
P 2 = 0.0484 (13)
The transit depth (δ) is calculated as:
δ=1− 0.9990/1 = 0.001 (14)
√δ= √0.001 ≈ 0.032 (15)

The planetary radius ($R_p$) relative to the stellar radius ($R_s$) is:
$R_p / R_s = \sqrt{\delta}$ (16)

$R_p = R_s \times \sqrt{\delta} = 0.8 \times 696,000 \times 0.032$ (17)
$R_p = 17,817.6$ km (18)
Converting this measurement to Earth radii ($R_\oplus$):

$17817.6/6371 \approx 2.8R_\oplus$ (19)

This indicates the planet is very close to Neptune-sized.

## Orbital Distance and Habitability:

Using Kepler's Third Law, the semi-major axis (a) is calculated:
a = (P^2× M )^1/3. (20)
a = (0.0484 × 0.8)^1/3= (0.03872)^1/3 (21)
a ≈ 0.338 AU (22)
Converting to kilometers:

a = 0.338 × 149.6 × 10^6 = 50.56 × 10^6 km (23)

This planet is about 3 times closer to its star than Earth is to the Sun.

The distance to the habitable zone ($d_{HZ}$) for this star is:
$d_{HZ} = \sqrt{0.40} \approx 0.632$ AU (24)

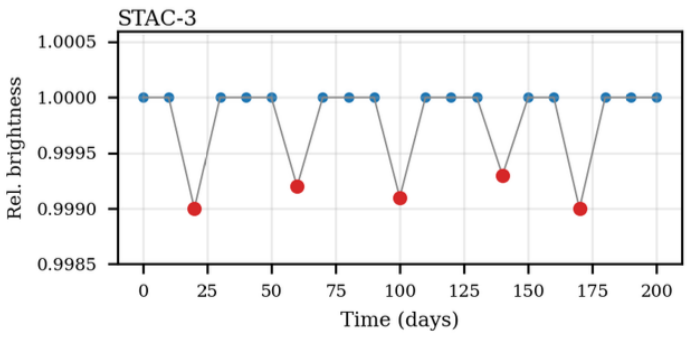
This planet sit at 0.338Au about 53% of that, might be hot or relatively hotter not too hot .

- It would better choice than 1 .

# STAC-3

Third host star properties :
- Mass: 0.50M⊙
- Radius: 0.50R⊙
- Luminosity: 0.08L⊙
- Surface temperature: 3800 K.

## Observations and Planetary Radius.




Brightness observations indicate dips at
The distance to the habitable zone (dHZ)
days 20, 60, 100, 140, and 170. While the gap is more accurately 38 days, for simplicity, it can be approximated as 40days.

The orbital period (P ) is therefore calculated as:

$P = 40/365.25 \text{ yr} \approx 0.11 \text{ yr}$ (25)

$P^2 = 0.0121$ (26)

Since the star shows variable dips, the average dipping is calculated as:

$$\text{Avg dipping} = (0.9990 + 0.9992 + 0.9993 + 0.9990)/4 \approx 0.999075 \approx 0.9991$$

The transit depth ($\delta$) is calculated as:

$\delta = 1.0000 - 0.9991/1.000 = 0.0009$ (28)

$\sqrt{\delta} = \sqrt{0.0009} = 0.03$ (29)

The planetary radius ($R_p$) relative to the stellar radius ($R_s$) is:

$R_p/R_s = \sqrt{\delta}$ (30)

$R_p = R_s \times \sqrt{\delta} = 0.5 \times 696,000 \times 0.03$ (31)

$R_p = 10,440 \text{ km}$ (32)

Converting this measurement to Earth radii ($R_\oplus$):

$10440/6371 \approx 1.64 R_\oplus$ (33)

This radius indicates the planet is super-Earth sized.

## Orbital Distance and Habitability

Using Kepler's Third Law, the semi-major axis (a) is calculated:

$a = (P^2 \times M)^{1/3}$ (34)

$a = (0.0121 \times 0.5)^{1/3} = (0.00605)^{1/3}$. (35)

$a \approx 0.182 \text{ AU}$ (36)

Converting to kilometers:

$a = 0.182 \times 149.6 \times 10^6 \approx 27.23 \times 10^6 \text{ km}$ (37)

This planet is about 5.5 times closer to its star than Earth is to the Sun.

$d_{HZ} = \sqrt{0.08} \approx 0.283 \text{ AU}$ (38)

This planet sit at 0.182AU , about 64% of that , relatively hotter .

Since it is closest to dHz and the closeness about 5.5 times , also radius is about the super earth sized.

- STAC-3 IS THE BEST CHOICE AND PROMISING.

## V. CONCLUSION

Each of the three stars hosts one transiting planet with periods of about 4, 80 and 40 days. STAC-3's Super-Earth, at 1.64Earth radii and 0.182 AU, is the most promising candidate for further study, and spectroscopy is the recommended next measurement.

## REFERENCES

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